

Unmanned Aerial Vehicles

Aeronaves Robotizadas

MEAer - Autumn Semestre – 2022/2023

<https://fenix.tecnico.ulisboa.pt/disciplinas/ARob/2022-2023/1-semester/pagina-inicial>

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Corpo Docente



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Responsável



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General Guidelines

- Contact hours :
 - Lectures and Exercises (T): once a week, 2h
Tuesday: 9h30 – 11h30
 - Laboratory (L): once a week, 1h30, total: 12
Wednesday: 10h30 – 12h
Wednesday: 12h30 – 14h
 - Attending hours (by appointment)
Schedule to be defined and indicated on Fenix page.
- Course website (Fenix page) :
<https://fenix.tecnico.ulisboa.pt/disciplinas/ARob/2021-2022/1-semester/pagina-inicial>

General Guidelines

- Laboratory Sessions:
 - Groups of 3, organized in 2 shifts (4 groups per shift)
 - Enrolment through Fenix from 21/9 until 25/9
 - Enrolment also possible in the lab on Wednesday 21/9 10h30
 - 3 Assignments (L1, L2, L3)
 - Handouts published in the course website under ‘Laboratories’
 - 4 lab sessions for each lab assignment:
 1. Modelling and identification of the drone
 2. Estimation of motion variables
 3. Motion control of the drone
 - 1 Report per assignment
 - Submission before the following lab session (check handout)
 - Use cover provided in Fenix tab ‘Laboratories’

General Guidelines

- Grading

- 1 Exam : 40%

Normal date (E1) : 26/1/2023, 13h

2nd chance date (E2) : 6/2/2023, 10h30

- 3 Lab assignments (L1, L2, L3) : 60%

- Final grade :

$$E = \max(E1, E2)$$

$$L = 0.2*L1 + 0.4*L2 + 0.4*L3$$

$$F = 0.4*E + 0.6*L$$

UAVs ?

- Example applications using UAVs
 - survey an area for S&R,
 - detect or track smuggling,
 - detect or track forest fires,
 - deliver a parcel
- What kind of UAV should we use ?
 - Airplane (fixed wing, flying wing)
 - Helicopter (or multi-rotor)
 - Autogiro (gyroplane or gyrocopter)
 - Bird (ornithopter)
 - Blimp (airship, balloon)

UAVs ?

Table 2.2 Flying principle comparison (1 = Bad, 3 = Good)

	Airplane	Helicopter	Bird	Autogiro	Blimp
Power cost	2	1	2	2	3
Control cost	2	1	1	2	3
Payload/volume	3	2	2	2	1
Maneuverability	2	3	3	2	1
Stationary flight	1	3	2	1	3
Low speed fly	1	3	2	2	3
Vulnerability	2	2	3	2	2
VTOL	1	3	2	1	3
Endurance	2	1	2	1	3
Miniaturization	2	3	3	2	1
Indoor usage	1	3	2	1	2
Total	19	25	24	18	25

UAVs ?

- Why a quadrotor drone ?
 - 4DOF actuation
 - Horizontal motion is omni-directional with 3 DOF
 - Plus vertical motion
 - VTOL capability
 - Simple and robust mechanics
 - Good controllability
 - Small, low cost sensors and control hardware
 - Usually 15min to 30min flight time
- And multi-rotors ?

UAVs ?

- Example applications using UAVs
 - survey an area for S&R or de-mining,
 - detect or track smuggling with an aircraft,
 - detect or track forest fires with an aircraft,
 - deliver a parcel
- So that the UAV should be able to :
 - stay in the air
 - navigate from one point to another
 - follow a road or desired path
 - avoid obstacles.

Syllabus

- Introduction to UAVs
 - Motivation, System Architecture, types of UAV configurations, applications.
- Modelling
 - Rigid body dynamic modelling.
 - Quadrotor dynamic modelling and design considerations.
- Control Systems Design
 - Quadrotor trajectory tracking control. Control design using root locus and loop shaping. Inner-outer loop control structure.
 - Linear systems and state space models.
 - Linear state feedback control and linear state observers. Duality and separation principle. LQR design example.

Syllabus

- **Sensors for UAVs**
 - Accelerometers. Rate gyros. Pressure sensors. Digital compasses. GNSS. Proximity sensors.
- **State Observers and Kalman Filtering**
 - Position, velocity, and attitude estimation including bias correction.
 - Use cases and connection to complementary filters
- **Nonlinear Systems**
 - Stability analysis and nonlinear control design.
 - Application to quadrotor trajectory tracking control.
- **Path following and path planning for UAVs**
 - Voronoi graphs, RRTs, Coverage algorithms
 - Dubins paths, waypoint transitions, optimization-based methods.

Schedule

UAVs 2022/2023			
planning on 220912			
week	Monday	Lecture	Lab
1	19-Set	Ch0 Intro Part I	<i>enroll</i>
2	26-Set	Ch0 Intro Part II	L1
3	03-Out	Ch1 Rigid Body	
4	10-Out	Ch2 Quadrotor Modeling	L1
5	17-Out	Ch3 Quadrotor Control Intro	L1
6	24-Out	Ch4 Modern Control Design	RL1
7	31-Out	<i>1st November</i>	L2
8	07-Nov	mid term	
9	14-Nov		
10	21-Nov	Ch5 Sensors Beard McLain	L2
11	28-Nov	Ch6 State Estimation Beard McLain	L2
12	05-Dez	<i>questions and answers</i>	RL2
13	12-Dez	Ch7 Intro NonlinearControl	L3
14	19-Dez	Ch8 Path Following	L3
		<i>Christmas holiday</i>	
15	02-Jan	Ch9 PathManager	L3
16	09-Jan	<i>questions and answers</i>	RL3

Exams : E1 26/1 & E2 6/2

Bibliography

Recommended:

- *Small Unmanned Aircraft: Theory and Practice*, Randal W. Beard & Timothy W. McLain, Princeton University Press, ISBN: 9781400840601, 2012.
- *Feedback Systems - An Introduction for Scientists and Engineers*, Karl Astrom and Richard Murray, online version at http://www.cds.caltech.edu/~murray/amwiki/index.php/Second_Edition

Additional:

- Flying Robots. In: Handbook of Robotics, B. Siciliano, O. Khatib, Eds. S. Leutenegger, C. Hürzeler, A. K. Stowers, K. Alexis, M. W. Achtelik, D. Lentink, P. Y. Oh, R. Siegwart, 2016. Springer.
- Handbook of Unmanned Aerial Vehicles, K. Valavanis, G. Vachtsevanos, Editors. 2015. Springer.
- Introduction to UAV Systems, P. Fahlstrom, T. Gleason, 2012. Wiley.

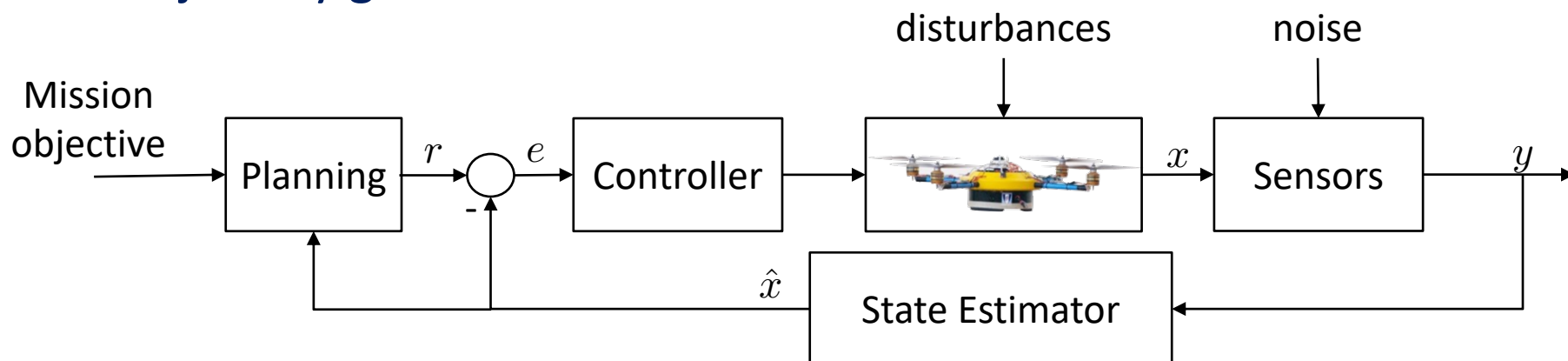
Introduction

General Topics

- Modelling (L1)
- Estimation (L2)
- Control (L3)
- Guidance / Path Planning / Trajectory generation

Control objective:

- Track a trajectory



Stabilization with wind disturbance rejection



Trajectory Tracking



Distributed Consensus-based Formation Control

